

CURRICULUM VITAE

Andrew, Chi-Chung Cheng

ORCID number: 0000-0003-4243-4192

EDUCATION

2021-2023	The Hong Kong Polytechnic University Research Postgraduate (Incomplete)
2021	The Hong Kong Polytechnic University BEng (1 st Hons) Electronic & Information Engineering

WORKING EXPERIENCES

From	To	Position
Sep 2023	Now	Department of Computer Science, Hong Kong Chu Hai College Research Assistant
Sep 2021	Sep 2023	Department of Electronic and Information Engineering, The Hong Kong Polytechnic University Teaching Assistant
Jul 2020	Aug 2020	Electrical and Mechanical Services Department Summer Intern

PROFESSIONAL POSITIONS/SERVICES

From	To	Position
Apr 2024	Mar 2026	Faculty of Dentistry, The University of Hong Kong Honorary Research Associate in Restorative Dental Sciences

AWARDS AND SCHOLARSHIP

2019/2020	The City Lions Club of Hong Kong Scholarship
2019/2020	The Dean's Honours List
2019	Best GPA Award
2019	Best Semester GPA Award
2018	Best Academic Improvement Award
2018	Best Semester GPA Award

RESEARCH INTERESTS

Artificial Intelligence, Deep Learning, Machine Learning, Interpretable Neural Networks, Image Segmentation

Technical Skills

Programming & Frameworks

Proficient	React.js, Django, Flask API, Python, HTML, CSS, JavaScript, PyQt5
Experienced	C++, C#, Unity, Java

Cloud & Machine Learning

Proficient	Google Cloud Platform (Cloud Run, Vertex AI), TensorFlow, PyTorch
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Languages

Native	Cantonese
Fluent	English, Mandarin

PROJECT EXPERIENCES

Hong Kong Chu Hai College

- Research on image segmentation for gum disease detection (2023-2025)

The Hong Kong Polytechnic University

- Research on interpretable neural networks via principal component analysis (2021-2023, as Research Postgraduate)
- Active noise cancellation with convolutional neural networks (2021, Undergraduate Final Year Project)

Electrical and Mechanical Services Department

- Developed IoT project using Arduino (ESP32) with mobile-controlled DHT11 sensors via Blynk cloud. (2020, as Summer Intern)

Publication and Patent List for Andrew, Chi-Chung Cheng

SUMMARY

My publications covering the following areas:

1. Image Segmentation.
2. Digital dentistry.
3. Artificial Intelligence.

PUBLICATIONS & PATENTS

1. Chi-Chung Cheng, Harris Sik-Ho Tsang, Richard Tai-Chiu Hsung, Yui-Lam Chan, Wai-Lun Lo and Walter Lam, "GumAgent: Towards an Accessible Gum Disease Detection Tool Leveraging Vision Language Model," *The 8th International Conference on Information Communication and Signal Processing* (ICICSP 2025)
2. Reinhard Chun Wang Chau, Andrew Chi Chung Cheng, In Meei Tew, Tien Hsin Cindy Chang, Hong Jin Tan, Khaing Myat Thu, Kaijing Mao, Colman McGrath, Wai-Lun Lo, Richard Tai-Chiu Hsung, Walter Yu Hang Lam. "Advancing Community-Based Oral Health: External Validation of an AI-Powered Smartphone App for Gingivitis Detection in Older Adults." 2024 TU-HKU-FJMU International Symposium on Oral Health Sciences. Sendai, Japan. Oct. 18-21, 2024.
3. Andrew Chi-Chung Cheng, Richard Tai-Chiu Hsung, Guan-Hua Li, In Meei Tew, Wai-Lun Lo, and Walter Yu Hang Lam, "Developing automated photographic detection of gum diseases using deep neural networks for mobile devices," 2024 6th International Conference on Software Engineering and Development (ICSED2024), May 29-31, 2024, Hong Kong, <https://doi.org/10.1145/3686614.3686621>.
4. Yu Hang LAM; Richard Tai Chiu HSUNG; Andrew Chi Chung CHENG; Guan-Hua LI; Wai-Lun LO; Chun Wang CHAU; and Khaing Myat THU, "Machine Learning-Based Image Processor For Intraoral Images", (2024) US patent application number 18/608,983. Filing Date: 03/19/2024.

OTHER GRANTS:

1. Reinhard Chun Wang Chau, Walter Yu Hang Lam, Richard Tai Chiu Hsung, Nelson Chun Yin Chau, Khaing Myat Thu, Andrew Chi Chung Cheng, Kaijing Mao; *Google Cloud x HKU Startup Connector*, Title: "Gingivitis Detection Solution", Duration: Jun 2024 - May 2026. Total Amount granted: USD250,000.